



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 AT3 TECHNICAL PRESENTATION

CSA0404 OPERATING SYSTEMS

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1. OPERATING SYSTEM IN A BANKING SYSTEM



Types of OS & OS Structure

- Multi-user OS – Supports thousands of users simultaneously.
- Multiprocessing OS – Executes multiple banking transactions at once.
- Time-sharing OS – Provides quick response to every customer.
- Layered OS Structure – Improves security and simplifies maintenance.
- Benefit: Ensures reliable and efficient banking services.



ROLE OF SYSTEM CALLS IN TRANSACTION PROCESSING



- Process Control – Creates and manages transaction processes (fork(), exec()).
- File Management – Accesses account records (open(), read(), write()).
- Memory Management – Allocates memory for transaction execution.
- Synchronization – Prevents data conflicts during simultaneous transactions.
- Outcome: Secure, fast, and accurate transaction processing.



2.OS FUNCTIONS IN A MULTI-USER LINUX SERVER



- CPU Scheduling – Shares processor time among users.
- Memory Management – Allocates memory efficiently.
- File Management – Organizes files and controls permissions.
- Device Management – Manages printers, disks, and other devices.
- Security – Authenticates users and protects resources.



FILE & DEVICE SYSTEM CALLS

USER-KERNEL INTERACTION



- File Calls: `open()`, `read()`, `write()`, `close()`
- Device Calls: `ioctl()`, `read()`, `write()`
- User applications request services through system calls.
- The kernel communicates with hardware and returns results.
- Ensures secure and controlled hardware access.



3.REAL-TIME EMBEDDED SYSTEM (TRAFFIC SIGNAL CONTROL)



- Uses a Real-Time Operating System (RTOS).
- Provides predictable and timely responses.
- Suitable for handling traffic sensors and signals.
- Components: Kernel, Scheduler, Timer, Device Drivers.
- Advantage: Reliable real-time performance.



SYSTEM CALLS FOR PROCESS CONTROL & TIMING



- Process Control: fork(), wait(), exit()
- Timing Functions: sleep(), alarm()
- Timer interrupts ensure accurate signal changes.
- Supports emergency vehicle priority.
- Improves traffic flow and road safety.



4.DISTRIBUTED OPERATING SYSTEM



- Connects multiple computers as one system.
- Supports resource sharing across locations.
- Provides load balancing and fault tolerance.
- Scales easily as the network grows.
- Common in cloud computing and enterprise systems.



ARCHITECTURE & COMMUNICATION



- Architecture: Client – Server – Network – Distributed Kernel.
- Communication through message passing.
- System Calls: socket(), connect(), send(), recv()
- Coordinates processes running on different machines.
- Ensures efficient communication and synchronization.



5.KEY OPERATING SYSTEM FUNCTIONS



- Process Scheduling – Manages CPU execution.
- Memory Allocation – Assigns RAM to programs.
- File Management – Stores and retrieves data.
- I/O Management – Controls input and output devices.
- Security – Protects system resources and users.



INTERACTION BETWEEN OS COMPONENTS



Flow:

Application → System Calls → Kernel → Hardware

Examples of System Calls:

Process: fork(), wait()

Memory: mmap(), brk()

File: open(), read(), write()

I/O: ioctl()

Conclusion: The OS coordinates all components to provide efficient, secure, and reliable computing.

THANK YOU!!!

